

Innodisk Firmware Upgrade - Download Microcode User Guide

Download Microcode Procedure:

1. Confirm the SSD name type
SSD model is 3SE4/3ME4 must be use A_COMBO_L17606.bin
SSD model is 3IE4 must be use A_COMBO_L17606i.bin
2. Copy binary file “XXXX.bin” to Home places

```

root@fae123-H67MA-USB3-B3: /home/fae123
fae123@fae123-H67MA-USB3-B3:~$ sudo su
[sudo] password for fae123:
root@fae123-H67MA-USB3-B3: /home/fae123# ls
Desktop    Downloads    Music        Public        Videos
Documents  examples.desktop  Pictures    Templates
root@fae123-H67MA-USB3-B3: /home/fae123# ls
A_COMBO_L17606.bin  Desktop    Downloads    Music        Public        Videos
A_COMBO_L17606i.bin Documents  examples.desktop  Pictures    Templates
root@fae123-H67MA-USB3-B3: /home/fae123#

```

3. Use the root privilege to execute the hdparm to do the download micro code
4. Type “**hdparm -i dev/sda**” to check current disk information

```

root@fae123-H67MA-USB3-B3: /home/fae123# hdparm -i /dev/sda
/dev/sda:
Model=2.5" SATA SSD 3ME4, FwRev=L16C28, SerialNo=20170605AAFF3A000001
Config={ HardSect NotMFM HdSw>15uSec Fixed DTR>10Mbps }
RawCHS=16383/16/63, TrkSize=0, SectSize=0, ECCbytes=0
BuffType=unknown, BuffSize=unknown, MaxMultSect=1, MultSect=1
CurCHS=16383/16/63, CurSects=16514064, LBA=yes, LBAsects=500118192
IORDY=on/off, tPIO={min:120,w/IORDY:120}, tDMA={min:120,rec:120}
PIO modes: pio0 pio3 pio4
DMA modes: mdma0 mdma1 mdma2
UDMA modes: udma0 udma1 udma2 udma3 udma4 udma5 *udma6
AdvancedPM=no WriteCache=enabled
Drive conforms to: unknown: ATA/ATAPI-3,4,5,6,7

* signifies the current active mode

```

5. Type “**sudo hdparm --yes-i-know-what-i-am-doing --please-destroy-my-drive --fwdownload A_COMBO_L17606.bin /dev/sda**” to upgrade firmware without losing data

```

root@fae123-H67MA-USB3-B3: /home/fae123# sudo hdparm --yes-i-know-what-i-am-doing
--please-destroy-my-drive --fwdownload A_COMBO_L17606.bin /dev/sda

```

6. After seeing “.....done !!”

```
/dev/sda:
fwdownload: xfer_mode=3 min=8 max=1024 size=4096
.....
..... Done.
```

7. Type “**smartctl -a /dev/sda**” check FW version

3SE4/3ME4 is “**L17606**”

3IE4 is “**L17606i**”

it is recommended to power off and re-power on

```
root@fae123-H67MA-USB3-B3:/home/fae123# smartctl -a /dev/sda
smartctl 6.2 2013-07-26 r3841 [x86_64-linux-3.13.0-32-generic] (local build)
Copyright (C) 2002-13, Bruce Allen, Christian Franke, www.smartmontools.org

=== START OF INFORMATION SECTION ===
Device Model:          2.5" SATA SSD 3ME4
Serial Number:         20170605AAFF3A000001
LU WWN Device Id: 5 02b2a2 01d1c1b1a
Firmware Version: L17606
User Capacity:         256,060,514,304 bytes [256 GB]
Sector Size:           512 bytes logical/physical
Rotation Rate:         Solid State Device
Device is:             Not in smartctl database [for details use: -P showall]
ATA Version is:        ACS-3 (unknown minor revision code: 0x011b)
SATA Version is:       SATA >3.1, 6.0 Gb/s (current: 6.0 Gb/s)
Local Time is:         Tue Jun  6 21:30:42 2017 EDT
SMART support is:      Available - device has SMART capability.
SMART support is:      Enabled

=== START OF READ SMART DATA SECTION ===
SMART overall-health self-assessment test result: PASSED
```

8. Firmware upgrade successfully after re-power on